Quantitative X-ray fluorescence mapping under white-beam excitation on BM32: the processing chain, illustrated on an (In,Ga)As/Si film

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Technical note – `pyXRT` repository, `docs/xrf_quantification` – October 15, 2026

Abstract

Laue microdiffraction maps on BM32 record an energy-dispersive fluorescence spectrum at every pixel, but the excitation is the full polychromatic spectrum of a bending magnet filtered by Ir mirrors, Be windows and several metres of air. Turning those spectra into compositions therefore needs every step of a fundamental-parameter analysis to be done with a white beam in mind. This note describes the complete chain implemented in the `pyXRT` modules `xrf_beamline`, `xrf_fit` and `xrf_matrix`: normalisation and counting statistics; the incident spectrum; the silicon-drift detector (resolution, line shape, efficiency, Si escape and pulse pile-up); flux-weighted line strengths; primary absorption and secondary fluorescence in a stack of layers; the constrained fit of the map-averaged spectrum; the self-consistent composition of a buried layer; a per-pixel linear solve; and the tracking of detector-resolution changes during the scan. Each step is illustrated on an 81×81-pixel map of a 50 nm InP / 1.9 µm (In,Ga)As / Si sample. The incident spectrum is computed for the apertures of the scan. Slit s3 accepts only 5.24 µrad vertically, a small fraction of the vertical width of the emission, so the sample sees the harder on-axis spectrum. This alone raises the Ga/As ratio by 4.5 % compared with a vertically integrated source. The Ga/As atom ratio of the film comes out as 0.5040 from the average spectrum and 0.5032 as the median of the map, with a pixel-to-pixel spread of 0.0052 of which 0.0044 is statistical. Over every plausible change of geometry, beam alignment, KB mirror angle, air path and layer thickness the ratio stays within 0.490–0.514. The In content, measured with the In L lines at 3.3 keV, is not under control: $(n_{\mathrm{Ga}} + n_{\mathrm{In}})/n_{\mathrm{As}} = 1.414$ instead of 1. The main reason is the 20 cm of air between sample and detector, which transmits only a few per cent of In Lα and whose length is not known to better than a few centimetres. We close with recommendations for the instrument and the analysis.

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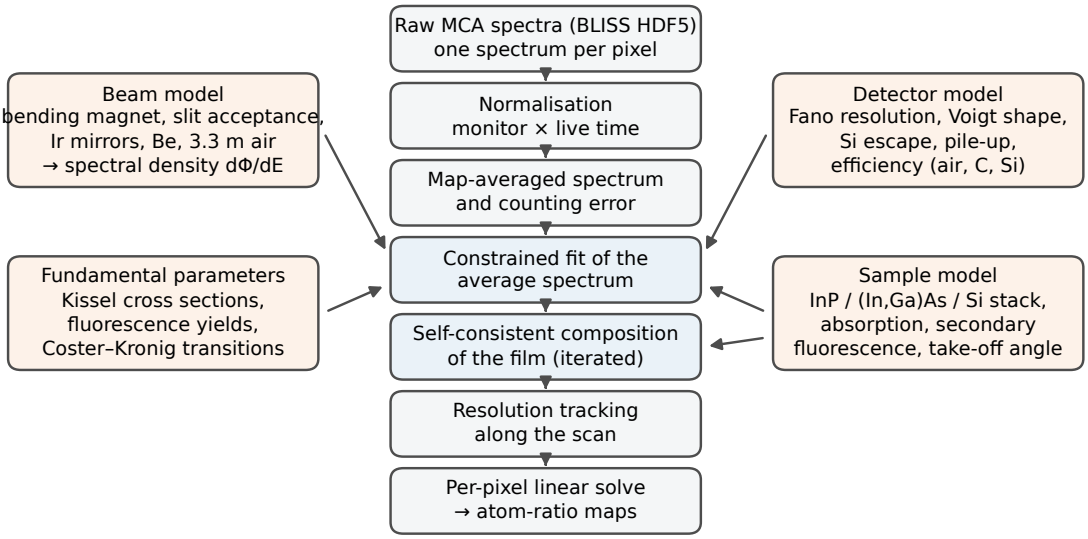


Figure 1: Processing chain. Side boxes are the physical models and data bases; the central column is the data flow. The two shaded steps are non-linear and run on the map-averaged spectrum; all per-pixel work is linear.

1 Introduction

White-beam (Laue) microdiffraction on BM32 [1] maps crystal orientation and strain with a sub-micrometre polychromatic beam. An energy-dispersive detector placed next to the sample records the X-ray fluorescence (XRF) at every pixel at no extra cost, so chemical maps come with every diffraction map. Turning these spectra into numbers raises problems that do not arise with a monochromatic beam:

1. every absorption edge of every element sits somewhere inside a 5–30 keV excitation band, so the relative strength of lines from different subshells and the relative sensitivity to different elements depend on the shape of that band;
2. the band itself is shaped by the source, four grazing-incidence mirrors, two Be windows and a measured 3.3 m incident air path, none of which is directly measured spectrally, and by slits that pass only a narrow vertical slice of the emission;
3. absorption of the exciting radiation and secondary fluorescence must be integrated over the band and over the depth of a layered sample;
4. the detector runs at about 100 kcps with a quarter of the time dead, so pile-up and any drift of its resolution are not negligible.

This note documents the chain that handles these effects, step by step. Each section states the physics, the equations as implemented, the approximations, and what the step changes on a real data set. All figures and all numbers are produced by the script `make_figures.py`, which sits next to this document and calls the same module functions as the analysis notebook `fluo_analysis_clean.ipynb`. Every number in the text is read from its output, `results.json`, so the document and the computation cannot drift apart.

The aim is clarity rather than brevity. Figure 1 gives the overview; readers interested only in using the code can skip to Section 13 and the recommendations in Section 15.

Table 1: BM32 elements between source and sample, settings of the InPoSi scan (positioners `vg1`, `hg1`, …, `ma1`, `mc1`, …), and full angular acceptance of each aperture referred to the source (Section 5.2). Slit gaps are read as millimetres. Mirror apertures are $L \sin\theta$. *KB grazing angles are not recorded; 3 mrad is assumed.

element	distance (m)	setting	aperture (V × H) (mm)	acceptance (µrad) vertical	acceptance (µrad) horizontal
s1	26.368		0.22 × 0.15	8.3	5.7
m1 (Ir, 1.2 m)	28.309	3.062 mrad, $R = 71.4$ km		129.8	–
m2 (Ir, 1.2 m)	31.732	2.551 mrad, $R = 145.7$ km		99.2	–
s2	35.370		10.00 × 10.00	303.6	282.8
s3	44.485		0.20 × 0.20	**5.2**	**4.5**
KB-V (Ir, 0.30 m)	44.645	3 mrad*		23.5	–
KB-H (Ir, 0.15 m)	44.880	3 mrad*		–	10.0
sample	45.0				

2 Experimental setup and data set

2.1 Beamline and geometry

The source is the 0.86 T bending magnet of ESRF-EBS (6.04 GeV); the map was recorded in 16 bunch mode at 76.5 mA. Table 1 lists the elements between source and sample with the settings recorded in the scan file. Primary slits s1 limit the fan accepted by the optics. Two Ir-coated mirrors m1 and m2, 1.2 m long, deflect vertically. They are bent (radii 71.4 km and 145.7 km), cut off the spectrum above about 28 keV and return the beam close to horizontal: the height difference of the two mirrors (21.1 mm) matches $2\theta_1$ times their separation (20.96 mm). Guard slits s2 remove scattered radiation. Slits s3 define the illumination of the Kirkpatrick–Baez (KB) pair, an elliptical Ir mirror focusing vertically and one focusing horizontally, which image the source on the sample. Two 500 µm Be windows and 3.3 m of air (measured) complete the excitation path. The KB grazing angles are not recorded in the scan files and are taken as 3 mrad. Section 5 shows that the apertures, through the vertical acceptance they impose, change the spectrum.

The sample is inclined so that both the incident beam and the detector axis make an angle of about 40° with its surface; these two angles are *assumed* in this note, not measured, and Section 13.2 shows what that costs.

2.2 Detector

The fluorescence detector is a KETEK VITUS H7 silicon-drift detector (SDD) with a 450 µm Si sensor and a 1 µm carbon (graphene) entrance window, read out by a XIA processor. Its window is about 20 cm from the sample, in air. The MCA has 2048 channels with the linear calibration

$$E\,[\mathrm{eV}] = 10.032\,c - 31.263, \tag{1}$$

so that it covers 0–20.5 keV. The processor records, per pixel, the live time (`xia_det0_live_time`), the real time (`xia_det0_elapsed_time`), the trigger (input) and event (output) count rates and the fractional dead time.

2.3 Sample

The sample is a 1.9 µm (In,Ga)As film grown on Si and capped with 50 nm of InP. The densities used are 4.81 g cm^{-3} (InP), 5.5 g cm^{-3} ((In,Ga)As, close to the mean of GaAs and InAs) and 2.33 g cm^{-3} (Si). The film composition is the unknown; the cap and the substrate are taken as known.

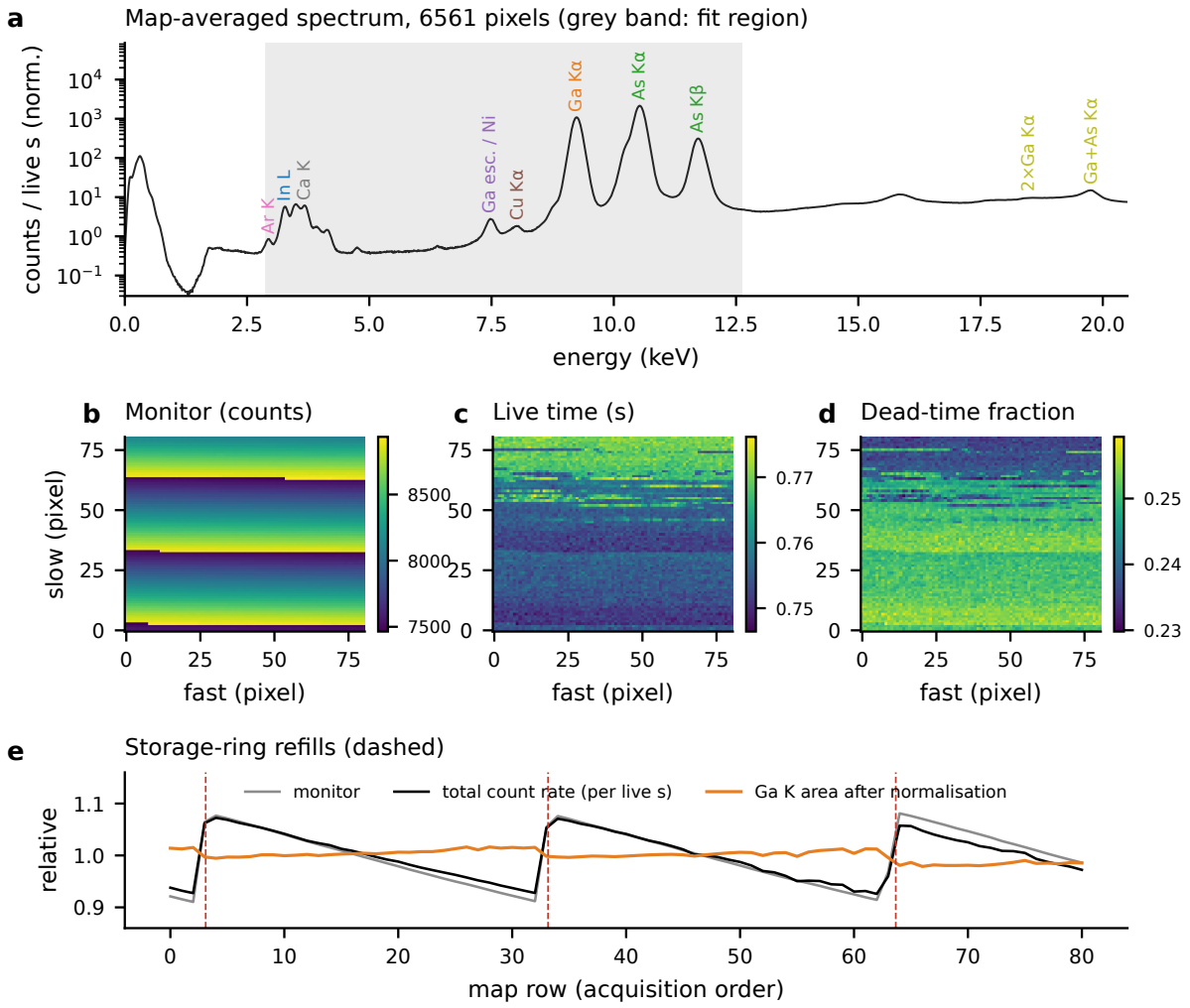


Figure 2: Data set. (a) Map-averaged, normalised spectrum (logarithmic scale); the grey band is the analysis region 2.9–12.6 keV. (b–d) Monitor, live time and dead-time fraction per pixel. (e) Row averages, in acquisition order, of the monitor, of the total count rate per live second, and of the Ga K area after normalisation by monitor and live time (Section 4); dashed lines mark the refills.

2.4 Data set

The map `InPoSi_v2_center.h5` is an 81×81 mesh (fast axis `xech`, slow axis `yech`, 0.25 µm steps) acquired in 1 s per point over 2.68 h. On average a pixel records 7.32×10^4 counts, 4.80×10^8 in total. The input count rate is 86.8–105.3 kcps (mean 96.6 kcps), the dead-time fraction 0.230–0.259 (median 0.250) and the live time 0.746–0.776 s for a real time of 1.008 s.

Figure 2 shows the map-averaged spectrum and the per-pixel normalisation counters. The spectrum is dominated by Ga K and As K. The In L lines appear between 3.1 and 4.2 keV on top of Ar K from air, with Ca K at 3.69 keV. A peak at 7.5 keV is mostly the Si escape peak of Ga Kα (Section 6.4), and Cu K appears at 8.05 keV. Above 12.6 keV there are sum peaks (Section 6.5) on a scattered background. The monitor (panel b) decays smoothly with the ring current and jumps by about 18 % at three storage-ring refills, about one hour apart, part-way through rows 3, 33 and 63 (panel e).

3 Overview of the processing chain

Figure 1 summarises the chain. Three physical models – the incident beam, the detector, and the sample – are combined with tabulated fundamental parameters to predict, for each element and shell, which lines are emitted, with which relative intensities, and how strongly per unit concentration. These predictions constrain the fit of the *map-averaged* spectrum. That fit is non-linear because it adjusts the detector resolution, and it is done once, on very high statistics. The fitted areas give the composition of the film self-consistently: the matrix correction depends on the composition, which is iterated. The line shapes are then frozen and every pixel is solved *linearly* for its line-group areas, which become atom-ratio maps through the converged sensitivities. One extra step checks whether the detector resolution stayed constant during the scan and accounts for it if not.

Notation. E is the incident photon energy and E_l the energy of fluorescence line l; c indexes MCA channels and i pixels; N_{ic} are raw counts; $\mu_k(E)$ is the total mass attenuation coefficient of layer k ($\mathrm{cm^2\,g^{-1}}$); $Q_l(E)$ the partial (line) fluorescence cross-section of the emitting element ($\mathrm{cm^2\,g^{-1}}$); $\varepsilon(E_l)$ the detection efficiency; W an atomic weight; w a mass fraction and n a number of atoms.

4 Normalisation and counting statistics

4.1 Per-pixel normalisation

The recorded counts of pixel i are converted into a rate per unit incident flux by dividing by the monitor (relative to its map mean) and by the live time:

$$y_{ic} = \frac{N_{ic}}{s_i}, \qquad s_i = \frac{m_i}{\bar{m}}\, t_i^{\mathrm{live}}. \tag{2}$$

Dividing by the live time corrects the dead-time losses, whatever their origin, because the XIA live time is the time during which the processor was able to accept a pulse. The correction is multiplicative and identical for all channels of a pixel. It therefore cancels exactly in any ratio of two areas from the same pixel, and so in atom-ratio maps. We checked this numerically: the Ga/As area ratio map obtained with and without Eq. (2) differs by at most 6.9×10^{-15} (relative). The normalisation matters for the absolute areas and for the average spectrum, in which pixels are combined.

4.2 Uncertainty of the average spectrum

The map-averaged spectrum is $\bar{y}_c = N^{-1} \sum_i y_{ic}$. Its *counting* uncertainty follows from Poisson statistics on the raw counts, $\mathrm{Var}(N_{ic}) = N_{ic}$:

$$\sigma_c = \frac{1}{N} \Big(\sum_i \frac{N_{ic}}{s_i^2} \Big)^{1/2}. \tag{3}$$

This is the weight used to fit $\bar{y}$. The standard error of the mean across pixels is a different quantity: it also contains real pixel-to-pixel variation, which is signal as far as the average is concerned. Here, at the peak, the two happen to be almost equal (9.5×10^{-6} and 9.5×10^{-6} of the peak height) because the map is close to homogeneous; on a heterogeneous map they can differ by an order of magnitude.

4.3 Monitor behaviour at storage-ring refills

Figure 2e tests the monitor. Before normalisation, the row-averaged count rate follows the monitor closely (correlation 0.983). Its row-to-row spread drops from 4.33 % to 1.14 % after division by the monitor. The correction is not perfect at the refills, however: the monitor jumps by 17.9 %, 17.6 % and 18.0 %, while the fluorescence count rate jumps by only 15.4 %, 15.1 % and 14.4 %. The normalised Ga K area therefore steps by −1.85 %, −1.83 % and −2.73 % (comparing the 200 pixels on either side), which is visible as horizontal bands in the area maps (Fig. 10a,b). The Ga/As ratio steps by only −0.06 %, −0.04 % and −0.11 %. The monitor over-responds: 14–20 % of its jump is not seen by the fluorescence. A dark offset or a non-linearity of the monitor, or a change of the focused beam after a refill (heat load on the optics), would each produce this; the present data cannot distinguish them. Absolute areas therefore carry a systematic error of about 2 % correlated with the refill cycle, whereas ratios are immune.

5 Incident white-beam spectrum

5.1 Source

The angular density of bending-magnet radiation, per horizontal and vertical angle and per 0.1 % bandwidth, is [2–4]

$$\left.\frac{\mathrm{d}^2\dot{N}}{\mathrm{d}\theta\,\mathrm{d}\psi}\right|_{0.1\%\,\mathrm{BW}} = 1.327\times10^{13}\,E_e^2[\mathrm{GeV}]\,I[\mathrm{A}]\,y^2(1+X^2)^2\left[K_{2/3}^2(\xi)+\frac{X^2}{1+X^2}K_{1/3}^2(\xi)\right] \qquad (4)$$

per mrad^2, with $y = E/E_c$, $X = \gamma\psi$, $\xi = \frac{y}{2}(1+X^2)^{3/2}$, $\gamma = 11820$ and the critical energy $E_c[\mathrm{keV}] = 0.665\,E_e^2[\mathrm{GeV}]\,B[\mathrm{T}] = 20.86\,\mathrm{keV}$. The first term is polarised in the orbit plane (horizontally), the second perpendicular to it. The emission is uniform in the horizontal angle θ but confined vertically to $|\psi| \sim 1/\gamma = 84.6\,\mu\mathrm{rad}$, and softer photons spread wider: the full width at half maximum of the vertical distribution is 271 µrad at 5 keV, 199 µrad at 10 keV, 139 µrad at 20 keV and 111 µrad at 30 keV (Fig. 3b). Integrated over all ψ, Eq. (4) gives the familiar vertically integrated flux

$$\left.\frac{\mathrm{d}\dot{N}}{\mathrm{d}\theta}\right|_{0.1\%\,\mathrm{BW}} = 2.457\times10^{13}\,E_e[\mathrm{GeV}]\,I[\mathrm{A}]\,G_1(y), \qquad G_1(y) = y\int_y^\infty K_{5/3}(x)\,\mathrm{d}x, \qquad (5)$$

whereas on axis ($X = 0$) its shape is $H_2(y) = y^2K_{2/3}^2(y/2)$, which is harder. Which of the two reaches the sample is set by the vertical acceptance of the beamline.

5.2 Beamline apertures and vertical acceptance

Each aperture of Table 1 limits the range of source angles that reaches the sample. To refer an aperture of size a at distance z to the source, a paraxial ray leaving the source with angle ψ is traced through the bent mirrors. A concave mirror of radius R at grazing angle θ acts on the divergence as a lens of focal length $f = R\sin\theta/2$:

$$h(z) = M(z)\,\psi, \qquad \Delta\psi_{\mathrm{acc}} = \frac{a}{M(z)}, \qquad (6)$$

where M grows linearly with distance and its slope $\mathrm{d}M/\mathrm{d}z$ drops by M/f at each bent mirror. The bending of m1 and m2 gives $f = 109\,\mathrm{m}$ and 186 m, which reduces the divergence downstream to 0.575 of its value at the source. The KB-V mirror therefore sees $M = 38.3\,\mathrm{m}$ instead of 44.6 m. Mirror apertures are their footprints, $L\sin\theta$. Table 1 and Fig. 3a give the acceptances. The limiting aperture in both planes is s3: 5.24 µrad vertically and 4.50 µrad horizontally. s1 comes next, and the mirrors and s2 never limit.

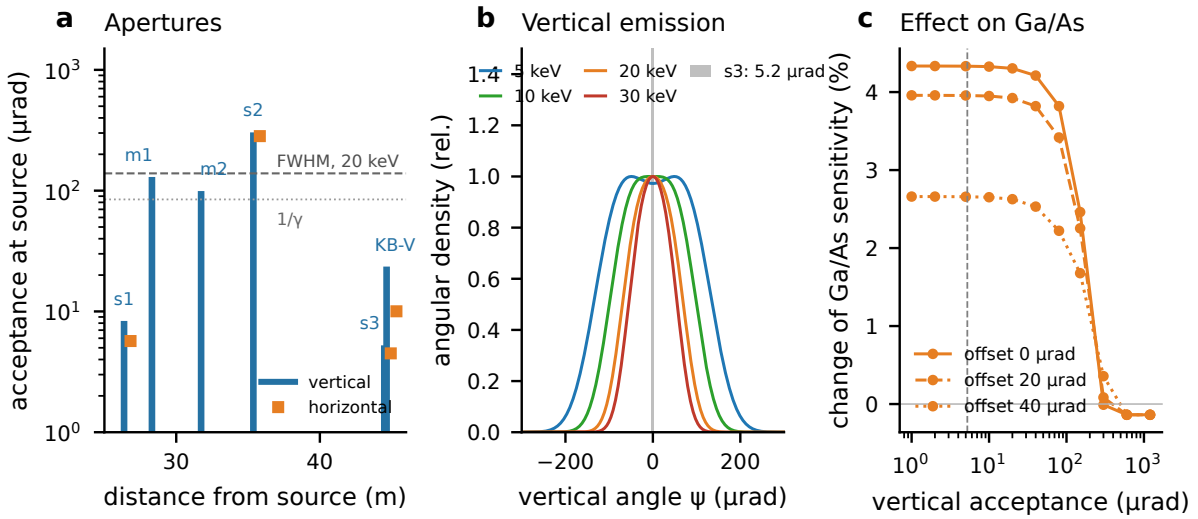


Figure 3: Vertical acceptance. (a) Full acceptance of each aperture referred to the source, vertical (bars) and horizontal (squares), compared with the width of the vertical emission at 20 keV and with $1/\gamma$. (b) Vertical angular distribution of the emission at four energies (each normalised) and the window accepted by s3. (c) Change of the bare Ga/As sensitivity ratio (K lines, beam and optics only) with respect to the vertically integrated source, versus full vertical acceptance, for the window centred on the orbit plane and offset by 20 and 40 µrad; the dashed line is the s3 acceptance.

The vertical acceptance is thus 21 to 52 times smaller than the vertical width of the emission (30 to 5 keV). The accepted spectrum is the on-axis spectrum, obtained by integrating Eq. (4) over $|\psi| \le \Delta\psi_{\mathrm{acc}}/2$, and not the vertically integrated one of Eq. (5). The horizontal acceptance only scales the flux, because the emission is uniform in θ. At the recorded current the model gives 4.7×10^8 photons/s per 0.1 % bandwidth at 20 keV on the sample. Fig. 3c shows how the Ga/As sensitivity ratio depends on the acceptance. It is constant to better than 0.1 % below about 20 µrad, so the exact gaps do not matter, and it relaxes to the vertically integrated value above 300 µrad. An error in the centring of the apertures on the orbit plane softens the spectrum: offsets of 10, 25 and 50 µrad change the Ga/As sensitivity by −0.09 %, −0.58 % and −2.67 %. The electron-beam divergence and the vertical source size of EBS are a few µrad and micrometres and are neglected. Because the accepted beam is on axis, it is linearly polarised in the horizontal plane. The KB-H mirror therefore reflects in p polarisation, which at 3 mrad differs from s by a negligible amount.

5.3 Optics and absorbers

Each mirror is described by its Fresnel reflectivity for s polarisation [5, 6]:

$$R(E,\theta) = \left|\frac{\sin\theta - q}{\sin\theta + q}\right|^2, \qquad q = \sqrt{n^2 - \cos^2\theta}, \qquad n = 1 - \delta(E) + \mathrm{i}\beta(E), \tag{7}$$

with the refractive index of Ir (22.56 g cm^{-3}) from xraylib [7, 8]. Interface roughness σ would multiply the reflection amplitude by the Névot–Croce factor $\exp(-2k_{z0}k_{z1}\sigma^2)$ [9], i.e. R by $\exp[-4k_0^2 \sin\theta\,\mathrm{Re}(q)\,\sigma^2]$. The roughness is unknown and set to zero here.

On the elliptical KB mirrors the grazing angle is not constant. Rays from the distant source arrive almost parallel but leave converging on the focus, so over an illuminated beam of size s the direction of the reflected ray changes by s/q, with q the distance to the focus, and the grazing angle by half of that:

$$\theta \in \theta_0 \pm \frac{s}{4q}, \qquad \bar{R}(E) = \frac{1}{2\Delta}\int_{\theta_0-\Delta}^{\theta_0+\Delta} R(E,\theta)\,\mathrm{d}\theta. \tag{8}$$

The beam passed by s3 is 0.20 mm high on KB-V ($q = 0.355$ m) and 0.20 mm wide on KB-H ($q = 0.12$ m). It illuminates about 67 mm and 67 mm of the mirrors, with $\Delta = 0.14$ mrad and 0.42 mrad. The averaging softens the high-energy cut-off of the KB pair between about 25 and 35 keV. It changes the Ga/As sensitivity by -0.15 %. The bending of m1 and m2 changes their grazing angle along the footprint by only $L/R \approx 0.02$ mrad, which is neglected.

Windows and air transmit $T = \exp[-(\mu/\rho)\rho t]$, with total attenuation coefficients and, for air, the density of an ideal gas of N, O, Ar and C at 20 °C and 1013.25 hPa. The spectrum at the sample is

$$\Phi(E) = \int_{\psi_0-\Delta\psi_{\mathrm{acc}}/2}^{\psi_0+\Delta\psi_{\mathrm{acc}}/2} \frac{\mathrm{d}^2\dot{N}}{\mathrm{d}\theta\,\mathrm{d}\psi}(E,\psi)\,\mathrm{d}\psi \prod_j T_j(E), \tag{9}$$

with $\psi_0 = 0$ for apertures centred on the beam, shown in Fig. 4a,b. The mirrors cut off above about 28 keV. The 3.3 m of air, not the Be windows, sets the low-energy end. The Ir L edges imprint three small steps between 11.2 and 13.4 keV. The band where Φ exceeds 10 % of its maximum is 9.0–29.4 keV (7.4–31.7 keV at 1 %). With a vertically integrated source it would be 8.6–29.1 keV. The mean photon energy is 18.56 keV instead of 17.89 keV.

5.4 From flux per bandwidth to spectral density

Eq. (9) is a flux per 0.1 % bandwidth, i.e. per interval $\mathrm{d}E = 10^{-3}E$. Any integral over photon energy needs the spectral density per unit energy,

$$\frac{\mathrm{d}\Phi}{\mathrm{d}E} = \frac{\Phi_{0.1\%}(E)}{10^{-3}\,E}, \tag{10}$$

which peaks at 18.1 keV instead of 21.6 keV (Fig. 4c); for a vertically integrated source it would peak at 16.6 keV. Forgetting this Jacobian over-weights high energies by a factor E. For the bare Ga/As sensitivity ratio $k = \int Q_{\mathrm{As}}\,\mathrm{d}\Phi\,W_{\mathrm{As}} / \int Q_{\mathrm{Ga}}\,\mathrm{d}\Phi\,W_{\mathrm{Ga}}$ (K lines, no detector or matrix) it gives 1.182 instead of 1.099, a bias of 7.5 % on the Ga/As ratio. For comparison, the vertically integrated source would give $k = 1.054$, i.e. the vertical acceptance moves k by 4.3 %.

6 Detector response

6.1 Energy calibration

The linear calibration of Eq. (1) is taken from the beamline and held fixed in the fit. Its stability is checked by the centroid analysis of Section 12: the Ga Kα centroid moves by 1.9 eV at most over the scan and shows no step. Its absolute accuracy has not been tested beyond the quality of the fit; refining gain and offset is one of the recommendations of Section 15.

6.2 Resolution and line shape

The Gaussian width of a line combines an electronic-noise term and the Fano-limited fluctuation of the number of electron–hole pairs [10, 11]:

$$\sigma^2(E) = \sigma_{\mathrm{n}}^2 + F\,w\,E, \qquad \mathrm{FWHM} = 2.3548\,\sigma, \tag{11}$$

with w the mean pair-creation energy. Each line is a Voigt profile, the convolution of this Gaussian with a Lorentzian of half-width γ that stands for the natural line width. σ_{n}, F and γ are common to all lines and fitted on the average spectrum. The implementation uses $w = 3.85$ eV; the measured value for Si at room temperature is $w = (3.66 \pm 0.03)$ eV [12]. Only the product Fw is determined by the data, so the choice only rescales the reported Fano factor (0.160 with 3.85 eV corresponds to 0.168 with 3.66 eV). The fit gives $\sigma_{\mathrm{n}} = 28.5$ eV, $F = 0.160$ and $\gamma = 11.9$ eV, i.e. 157 eV FWHM at Mn Kα and 190 eV at Ga Kα. That is broader than the 129 eV quoted at

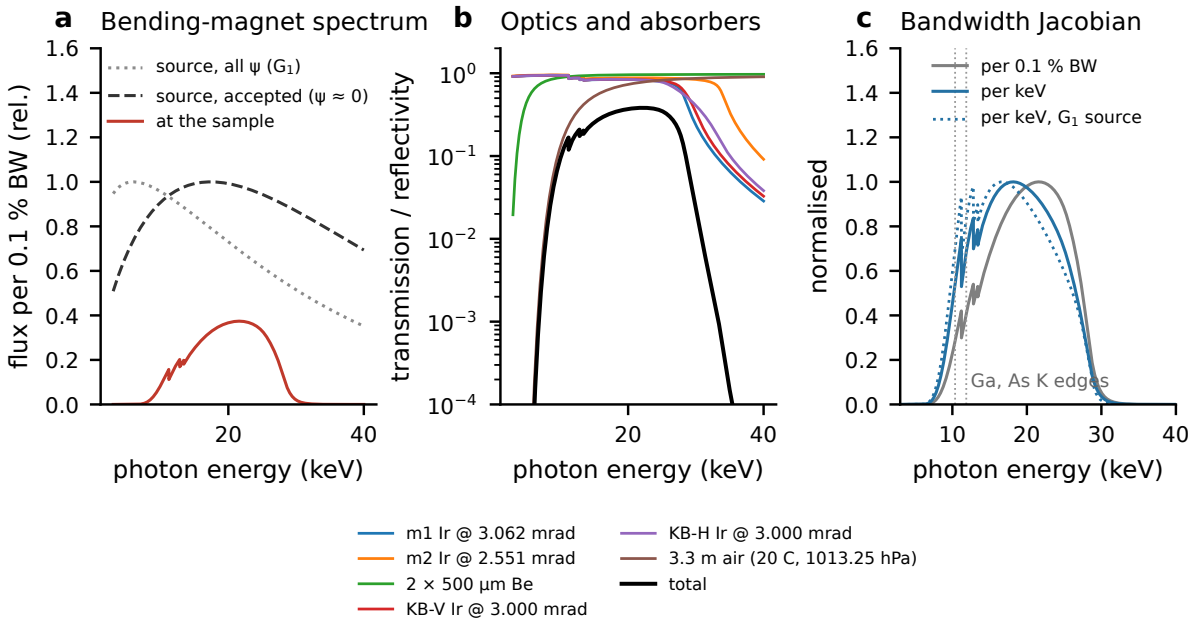


Figure 4: Incident spectrum. (a) Bending-magnet flux per 0.1 % bandwidth at the source, vertically integrated (G_1) and inside the accepted window, and after the optics, each normalised to its source maximum. (b) Transmission or reflectivity of each element (KB mirrors averaged over their angle spread) and their product. (c) The spectrum at the sample per 0.1 % bandwidth and per unit energy (Eq. 10), and per unit energy for a vertically integrated source, each normalised to its maximum; dotted lines mark the Ga and As K edges.

optimum shaping, as expected at 100 kcps, where a short peaking time is used. Over 3–12 keV the noise and Fano terms are strongly correlated, and their values should be read as an effective resolution law, not as detector physics. The Lorentzian width is larger than natural widths of K lines (a few eV). It partly absorbs a low-energy tail that the model does not contain (Section 9).

6.3 Detection efficiency

The probability that a photon emitted towards the detector is recorded in the full-energy peak or its escape peak is

$$\varepsilon(E_l) = \mathrm{e}^{-\mu_{\mathrm{air}}(E_l)\rho_{\mathrm{air}}d_{\mathrm{air}}}\ \mathrm{e}^{-\mu_{\mathrm{C}}(E_l)\rho_{\mathrm{C}}d_{\mathrm{C}}}\ \left[1-\mathrm{e}^{-\tau_{\mathrm{Si}}(E_l)\rho_{\mathrm{Si}}d_{\mathrm{Si}}}\right], \tag{12}$$

using total attenuation coefficients along the path (any interaction removes the photon) and the photoelectric coefficient τ in the sensor (a scattered photon does not deposit its full energy). No dead layer is assumed. Figure 5a shows the factors for $d_{\mathrm{air}} = 20$ cm, $d_{\mathrm{C}} = 1$ µm and $d_{\mathrm{Si}} = 450$ µm. The efficiency is 0.845 at Ga Kα_1, 0.853 at As Kα_1, 0.819 at As Kβ_1, but only 0.0356 at In Lα_1. The air path dominates that number: one more centimetre of air multiplies the In Lα_1 efficiency by 0.847 and the Ga Kα_1 efficiency by 0.992 (Fig. 5b). The local pressure matters too: 990 hPa instead of 1013 hPa raises the In Lα_1 efficiency by a factor 1.079. This single factor is the main uncertainty of any quantification based on soft lines (Section 13.2).

6.4 Silicon escape peaks

A photon absorbed in the Si K shell near the entrance of the sensor may be followed by a Si K X-ray that leaves the crystal; the event is then recorded at $E - E_X$. For normal incidence on a thick sensor the escape fraction is [13]

$$f_X(E) = \tfrac{1}{2}\,\omega_K\,\frac{\tau_K(E)}{\tau(E)}\,p_X\left[1-\frac{\mu(E_X)}{\tau(E)}\ln\!\left(1+\frac{\tau(E)}{\mu(E_X)}\right)\right], \tag{13}$$

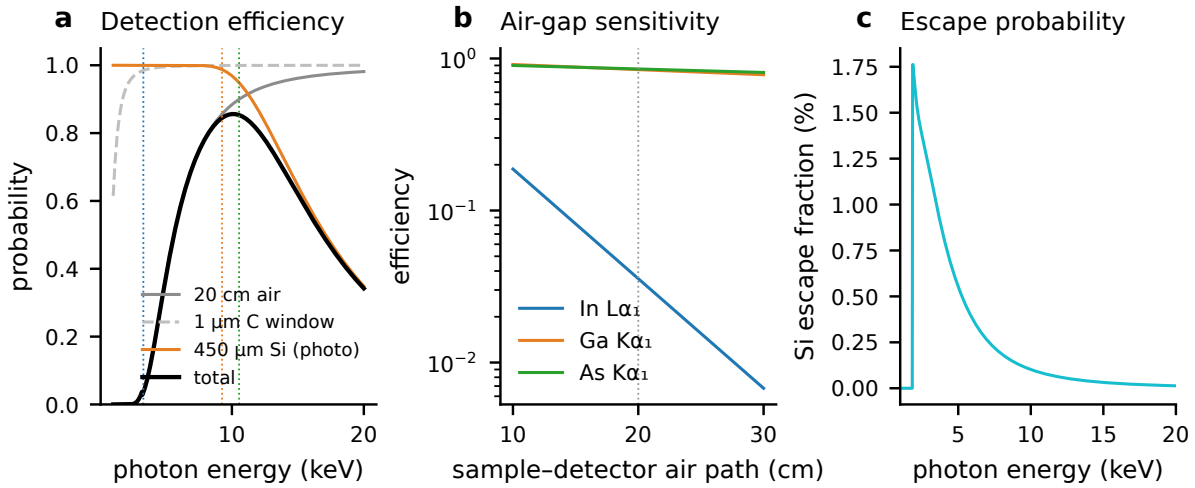


Figure 5: Detector. (a) Transmission of 20 cm of air and of the 1 µm carbon window, photo-absorption in 450 µm Si, and the resulting efficiency (Eq. 12); dotted lines mark In Lα_1, Ga Kα_1 and As Kα_1. (b) Efficiency versus sample–detector air path. (c) Total Si K escape fraction (Eq. 13).

with ω_K the Si K fluorescence yield, τ_K/τ the K-shell share of photo-absorption at E, p_X the fraction of Kα or Kβ among Si K emissions, $\tau(E)$ the photo-absorption coefficient of Si at E and $\mu(E_X)$ the attenuation of the escaping line. The fraction reaches 1.76 % just above the Si K edge and falls to 1.08 % at In Lα and 0.126 % at Ga Kα (Fig. 5c). Every line of the model carries its escape companions with this fraction and the resolution of the deposited energy $E - E_X$, with no extra free parameter. The group area then counts every photon absorbed in the sensor, consistently with Eq. (12).

The Ga Kα escape peak lies at 7.51 keV, 30 eV from Ni Kα (7.48 keV). Without escape peaks the fit attributes 30.4 counts/s to Ni K. With them, the Ni area drops to 12.6 counts/s (Fig. 6a). The remainder is either genuine Ni or an underestimate of the escape fraction. Eq. (13) ignores escape through the sides of a collimated sensor and the incomplete charge collection near its entrance window. The overall χ^2 does not improve with the escape peaks (1493 without, 1524 with): the misfit around 8.8 keV, where the As Kα escape adds to the unmodelled low-energy tail of Ga Kα, outweighs the gain at 7.5 keV. The escape term is kept on physical grounds.

6.5 Pulse pile-up

Two photons arriving within the pulse-pair resolving time τ_{pp} of the pile-up inspector are recorded as one event at the sum of their energies. For Poisson arrivals, and to first order, the pile-up spectrum of a pixel is [14]

$$N_i^{\mathrm{pp}}(E) = \frac{\tau_{\mathrm{pp}}}{t_i^{\mathrm{real}}}\,(x_i \otimes x_i)(E), \qquad (x \otimes x)(E) = \sum_{E_1+E_2=E} x(E_1)\,x(E_2), \tag{14}$$

where x_i is the measured spectrum of the pixel and the sum runs over ordered pairs. The self-convolution is computed per pixel by FFT over the full MCA range, with the calibration offset counted twice and the diagonal term made unbiased ($x_c(x_c-1)$). Pile-up goes as the square of the local rate, so the template for the average spectrum is the average of the per-pixel N_i^{pp}/s_i, not the self-convolution of the average. Pile-up removes the two parent events regardless of their energies, so it does not bias line ratios; its effect is the sum peaks, which other components of the model would otherwise absorb.

The main sum peaks (18.5 and 19.8 keV) lie above the analysis region, so τ_{pp} is fitted once over 2.9–20.3 keV (background order 5) and then held fixed. This gives $\tau_{\mathrm{pp}} = (0.1150 \pm 0.0002)$ µs (statistical) and lowers χ^2_ν of that extended fit from 1429 to 1295 (Fig. 6b). At the mean input

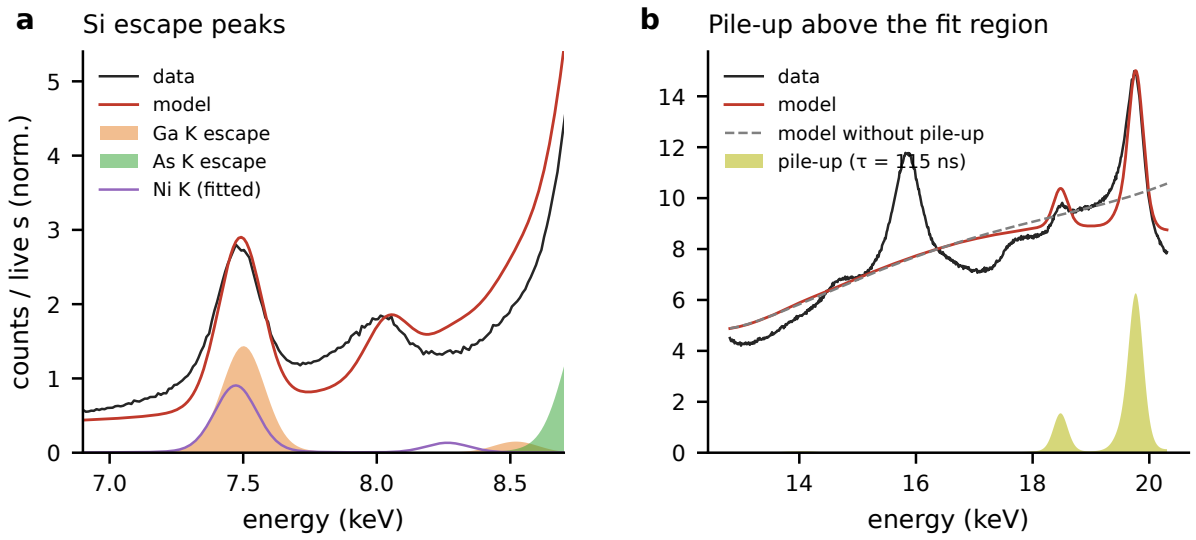


Figure 6: (a) Region of the Si escape peaks of Ga K and As K. The escape components (shaded) are tied to their parent lines; the purple line is the Ni K group still fitted. (b) Extended fit up to 20.3 keV with and without the pile-up term (Eq. 14); the shaded areas are the predicted 2×Ga Kα and Ga Kα+As Kα sum peaks for the fitted resolving time.

rate the probability that a given event is piled up is about 1.1 %. A single τ_{pp} describes the Ga+As sum peak well but overestimates the 2×Ga Kα peak, consistent with a resolving time that depends on the energies of the two pulses [14]. A broad peak at 15.8 keV, with weaker features near 14.5 and 17.5 keV, is not part of the model. Its energy matches Zr Kα (15.78 keV, with Kβ at 17.67 keV), but a Bragg reflection of the white beam by the single-crystal sample reaching the detector cannot be excluded. It lies outside the analysis region.

7 Fundamental parameters and line groups

7.1 Line fluorescence cross-sections

For each element and each shell family (K, L) the model contains every transition listed by xraylib under its IUPAC name (e.g. `KL3`, `L3M5`). Siegbahn aliases and composite macros (`KA`, `LA`) are excluded so that no intensity is counted twice. The emission of line l of element i per unit mass, for an incident photon of energy E, is the partial fluorescence cross-section

$$Q_{il}(E) = \Big[\tau_{L_3}(E) + f_{23}\tau_{L_2}(E) + (f_{13} + f_{12}f_{23})\tau_{L_1}(E)\Big]\,\omega_{L_3}\,p_l \qquad \text{(for an } L_3 \text{ line)}, \tag{15}$$

with subshell photo-ionisation cross-sections τ (Kissel), Coster–Kronig transfer probabilities f, fluorescence yields ω and radiative branching ratios p_l (`CS_FluorLine_Kissel` [7, 8, 15–17]). K lines have the same form without the transfer terms. Q is comparable across subshells, which radiative rates alone are not. That matters for L lines, which originate from three subshells with different edges.

7.2 Flux-weighted line strengths

Under white-beam excitation the strength of line l, before matrix effects, is

$$s_{il} = \varepsilon(E_l) \int Q_{il}(E)\,\frac{\mathrm{d}\Phi}{\mathrm{d}E}\,\mathrm{d}E, \tag{16}$$

integrated with the trapezoidal rule on a 10 eV grid from 3 to 40 keV. The trapezoidal rule cannot overshoot at the edge discontinuity. The lines of a group (element, family) whose strength exceeds

Table 2: Predicted relative intensities (to the reference line of the group) and reduced χ^2 of the average-spectrum fit at increasing levels of physics. All fits include escape and pile-up.

level	In L, relative to L_3M_5 ($L\alpha_1$)			As K, relative to KL_3		χ^2_ν
	L_2M_4 ($L\beta_1$)	L_3N_5 ($L\beta_2$)	L_1M_3 ($L\beta_3$)	KL_2	KM_3	
$E_{\mathrm{eff}} = 20\,\mathrm{keV}$	0.446	0.110	0.159	0.5153	0.1467	1611
white beam	0.473	0.110	0.127	0.5153	0.1467	1617
+ detector efficiency	0.799	0.297	0.259	0.5156	0.1408	1649
+ sample (stack)	0.852	0.335	0.278	0.5152	0.1436	1531

0.5 % of the strongest are kept, and each gets a fixed relative intensity $r_l = s_{il}/s_{i,\mathrm{ref}}$. For the stack of Section 8, s is replaced by the sensitivity of Eq. (21).

Table 2 shows how the successive physics levels change the predicted line pattern. An effective excitation energy of 20 keV gives In $L\beta_1/L\alpha_1 = 0.446$. Weighting by the white beam changes the share of the three L subshells, and the detector efficiency, which is much higher at $L\beta_1$ (3.49 keV) than at $L\alpha_1$ (3.29 keV), nearly doubles the ratio. Self-absorption in the film raises it further, because the film absorbs $L\alpha_1$ more strongly than $L\beta_1$.

8 Sample model: absorption and secondary fluorescence in a layer stack

8.1 Primary fluorescence

Layers are numbered from the surface, $k = 0, 1, \ldots$; layer k has mass thickness $M_k = \rho_k t_k$ (infinite for a bulk last layer). With incidence and take-off angles ψ_{in} and ψ_{out} measured from the surface, define

$$a_k(E) = \frac{\mu_k(E)}{\sin\psi_{\mathrm{in}}}, \quad b_k = \frac{\mu_k(E_l)}{\sin\psi_{\mathrm{out}}}, \quad F(c;M) = \frac{1-\mathrm{e}^{-cM}}{c}, \quad G(x,y;M) = \frac{\mathrm{e}^{-yM}-\mathrm{e}^{-xM}}{x-y}. \tag{17}$$

The primary sensitivity of line l of element i in layer k, per unit mass fraction, is [18–20]

$$P^k_{il} = T^k_{\mathrm{out}} \int \frac{\mathrm{d}\Phi}{\mathrm{d}E}\,\frac{Q_{il}(E)}{\sin\psi_{\mathrm{in}}}\,T^k_{\mathrm{in}}(E)\,F\big(a_k(E)+b_k;M_k\big)\,\mathrm{d}E, \tag{18}$$

with the overlayer transmissions $T^k_{\mathrm{in}} = \exp(-\sum_{c<k} a_c M_c)$ and $T^k_{\mathrm{out}} = \exp(-\sum_{c<k} b_c M_c)$. The ratio of P^k to its value without any absorption ($F \to M_k$, no overlayers) is the absorption factor A.

8.2 Secondary fluorescence

Line j of element e, excited by the beam in layer α, is emitted isotropically and can be absorbed by element i in layer k, which then emits line l. Writing $u = \cos\vartheta$ for the direction of the secondary photon and $\kappa_c = \mu_c(E_j)/u$, the depth integrals are closed-form for every u, and [19–21]

$$S^k_{il} = T^k_{\mathrm{out}} \sum_{\alpha,\,e,\,j} \tfrac{1}{2}\,Q_{il}(E_j)\,w^\alpha_e \int \frac{\mathrm{d}\Phi}{\mathrm{d}E}\,\frac{Q_{ej}(E)}{\sin\psi_{\mathrm{in}}}\,T^\alpha_{\mathrm{in}}(E)\int_0^1 K_{\alpha k}(E,u)\,\frac{\mathrm{d}u}{u}\,\mathrm{d}E, \tag{19}$$

with the kernels

$$\begin{aligned}
K_{kk} &= \frac{F(a_k+\kappa_k)-F(a_k+b_k)}{b_k-\kappa_k} + \frac{F(b_k+\kappa_k)-F(a_k+b_k)}{a_k-\kappa_k} && \text{(same layer)},\\
K_{\alpha k} &= G(a_\alpha,\kappa_\alpha;M_\alpha)\,\mathrm{e}^{-\sum_{\alpha<c<k}\kappa_c M_c}\,F(b_k+\kappa_k;M_k) && (\alpha<k,\ \text{downward}),\\
K_{\alpha k} &= F(a_\alpha+\kappa_\alpha;M_\alpha)\,\mathrm{e}^{-\sum_{k<c<\alpha}\kappa_c M_c}\,G(b_k,\kappa_k;M_k) && (\alpha>k,\ \text{upward}).
\end{aligned} \tag{20}$$

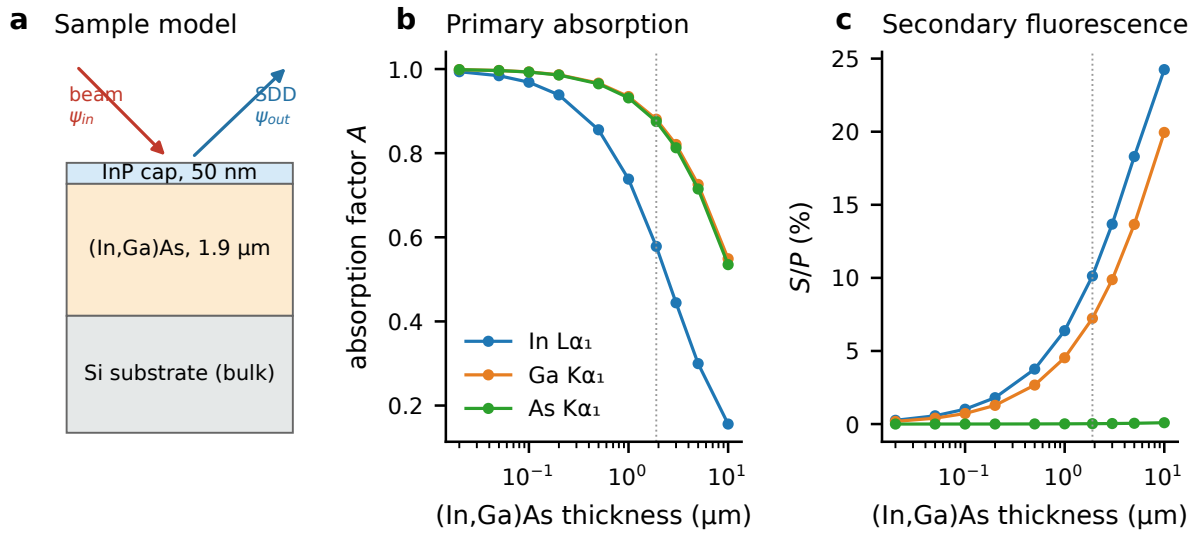


Figure 7: Sample model. (a) Stack and geometry. (b) Absorption factor (primary intensity relative to a non-absorbing layer) and (c) secondary-to-primary ratio of In Lα_1, Ga Kα_1 and As Kα_1 for an $In_{0.5}Ga_{0.5}As$ layer ($5.5\,\mathrm{g\,cm^{-3}}$) of varying thickness, $\psi_{\mathrm{in}} = \psi_{\mathrm{out}} = 40°$; the dotted line marks 1.9 µm.

The same-layer kernel is the closed form of $\int_0^M \int_0^M \mathrm{e}^{-am-bm'-\kappa|m-m'|}\mathrm{d}m\,\mathrm{d}m'$; its removable singularities at $b = \kappa$ and $a = \kappa$ are handled by the derivative of F. The u integral is done on a logarithmic grid of 160 points, which resolves the grazing paths that dominate in thin films. The energy integral uses a coarser 40 eV grid. For a single bulk layer Eq. (19) reduces to the classical Shiraiwa–Fujino expression. Tertiary fluorescence, excitation by scattered radiation and interface roughness are not included.

The implementation was checked against brute-force numerical integration of the double depth integral (agreement to 1×10^{-6}), against the classical bulk expression (4×10^{-4}) and against an independent estimate of the Ga K enhancement in a 200 nm film (1.20 % against 1.19 %). Splitting a layer into two identical sublayers changes nothing to within 10^{-10}.

8.3 Recorded intensity

The intensity of line l recorded from the whole stack is

$$I_{il} \propto \varepsilon(E_l) \sum_k w_i^k \left(P_{il}^k + S_{il}^k\right). \tag{21}$$

Figure 7 shows how absorption and secondary fluorescence evolve with the thickness of an $In_{0.5}Ga_{0.5}As$ layer alone. At 1.9 µm the absorption factor is 0.881 for Ga Kα but 0.578 for In Lα, and the secondary fluorescence adds 7.2 % to Ga Kα (excited by As K) and 10.1 % to In Lα (excited by As K and Ga K). As K is not enhanced. Both effects grow without saturation up to 10 µm (Ga 19.9 %, In 24.2 %): the film is not thin for any of the lines of interest.

9 Fit of the map-averaged spectrum

9.1 Model

Over the analysis region (channels 292–1261, 2.9–12.6 keV) the model is

$$\hat{y}(c) = \sum_{k=0}^{K} \beta_k u^k + \tau_{\mathrm{pp}}\,\bar{N}^{\mathrm{pp}}(c) + \sum_g A_g \sum_{l \in g} r_l \Big[\big(1 - \textstyle\sum_X f_X\big) V(c - c_l; \sigma_l, \gamma) + \sum_X f_X\, V(c - c_{l,X}; \sigma_{l,X}, \gamma)\Big], \tag{22}$$

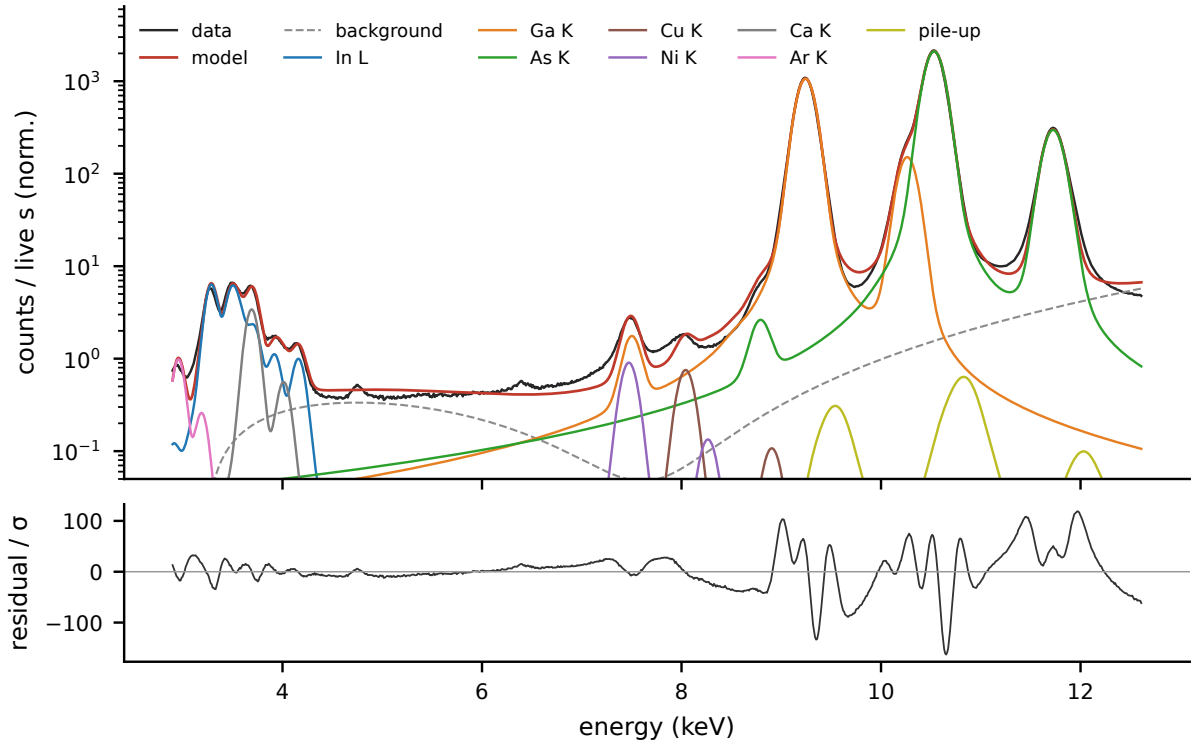


Figure 8: Fit of the map-averaged spectrum with the full model (white beam, detector efficiency, stack with secondary fluorescence, escape, fixed pile-up). Coloured lines are the line groups including their escape peaks; the lower panel is the residual in units of the counting error.

with $u \in [-1, 1]$ the scaled channel, a polynomial background of order $K = 3$, V a unit-area Voigt profile in channel units, c_l the channel of line l from Eq. (1), σ_l from Eq. (11), and A_g the area of the reference line of group g (escape included). The relative intensities r_l and escape fractions f_X are fixed by Sections 6–8; the pile-up term $\bar{N}^{\mathrm{pp}}$ is the average template of Eq. (14) with τ_{pp} fixed. The free parameters are the background coefficients, one area per group (In L, Ga K, As K, Cu K, Ni K, Ca K, Ar K), and σ_{n}, F, γ: 14 parameters for 39 lines in 7 groups. The fit is a weighted least-squares problem with the counting errors of Eq. (3) (trust-region reflective algorithm with bounds, `scipy.optimize.curve_fit`).

9.2 Result

Figure 8 shows the fit. The r.m.s. residual is 0.42 % of the maximum, but $\chi^2_\nu = 1524$ over 969 channels. With 4.80×10^8 counts in the average, the counting errors are tiny and χ^2 is set by systematic line-shape deviations. The residuals show their structure: excess counts on the low-energy side of each strong peak (incomplete charge collection, i.e. a tail and shelf that a Voigt profile does not have [22]) and the corresponding overshoot of the fitted Lorentzian wings between the peaks. The fitted areas (counts/s, per unit relative monitor) are In L 89.32±0.11, Ga K 16074±1.4, As K 33590±2.1, Cu K 10.76, Ni K 12.57, Ca K 37.42 and Ar K 10.13; the quoted errors are statistical only. Successive physics levels change χ^2_ν only moderately (Table 2). The white-beam and efficiency levels alone slightly increase it, and the full sample model gives the lowest value. This suggests that the predicted In L pattern becomes consistent with the data only once self-absorption is included.

Table 3: Matrix terms of the converged quantification for the reference line of each group in the (In,Ga)As film: absorption factor A, secondary-to-primary ratio S/P, fraction of the modelled intensity emitted by the other layers, and transmission of the 50 nm InP cap.

line	A	S/P (%)	other layers (%)	cap transmission
In Lα_1	0.583	7.82	5.9	0.980
Ga Kα_1	0.867	5.60	0	0.995
As Kα_1	0.873	0.036	0	0.996

10 Self-consistent quantification of a buried layer

10.1 Single layer

For a homogeneous sample, the reference-line area of group g is $A_g \propto w_g s_g$, with $s_g = \varepsilon(P + S)$ the sensitivity per unit mass fraction. With $w = nW/\sum nW$, the atom ratio of two elements is

$$\frac{n_a}{n_b} = \frac{A_a}{A_b}\,\frac{s_b\,W_b}{s_a\,W_a}. \tag{23}$$

Because s depends on the composition through absorption and secondary fluorescence, Eq. (23) is iterated: start from a nominal composition, compute the ratios, rebuild the layer, recompute s, until the ratios change by less than 10^{-6} [23].

10.2 Target layer in a stack

When other layers emit the same lines (here In L from the InP cap), the measured intensity is

$$A_g \propto w_g^t s_g^t + C_g, \qquad C_g = \sum_{k \neq t} w_g^k s_g^k, \tag{24}$$

where t is the target layer and the other layers have known composition. Relative to the reference group r (As K),

$$w_i^t = \frac{(A_i/A_r)\,(w_r^t s_r^t + C_r) - C_i}{s_i^t}, \tag{25}$$

and the atom ratios follow from the mass fractions. The absolute mass fraction of the reference enters once $C \neq 0$, so unmeasured elements of the target layer must be fixed or neglected. Density and thicknesses are held fixed; the iteration converged in 4 steps.

10.3 Result on the average spectrum

The film composition per As atom is

$$n_{\mathrm{Ga}}/n_{\mathrm{As}} = 0.5040, \qquad n_{\mathrm{In}}/n_{\mathrm{As}} = 0.910, \qquad (n_{\mathrm{Ga}} + n_{\mathrm{In}})/n_{\mathrm{As}} = 1.414. \tag{26}$$

Table 3 lists the matrix terms. The Ga/As ratio is close to the raw area ratio ($k = 1.0533$) because the absorption of Ga Kα and As Kα is similar (0.867 and 0.873) and the 5.60 % secondary enhancement of Ga by As K (KL_3 3.26 %, KL_2 1.69 %, KM_3 0.40 %) is compensated by the Ga/As cross-section ratio. Without the sample model (white beam and efficiency only) the same areas give $n_{\mathrm{Ga}}/n_{\mathrm{As}} = 0.5275$, i.e. the sample model changes the result by -4.4 %.

The InP cap matters only for In: it emits 5.9 % of the modelled In L intensity and transmits 0.980 of In Lα and more than 0.995 of Ga and As Kα. Ignoring the cap gives $n_{\mathrm{In}}/n_{\mathrm{As}} = 0.949$ (closure 1.453) and leaves Ga/As unchanged (0.5041). The closure deficit is therefore not a cap effect: even a 500 nm cap would leave it at 1.113 (against 1.414 at 50 nm).

11 Per-pixel linear solve

Once the resolution parameters and τ_{pp} are fixed, Eq. (22) is linear in the background coefficients and the group areas. For pixel i, with design matrix D (one column per background term and per group profile), the weighted least-squares solution and its covariance are

$$\hat{\mathbf{a}}_i = (D^{\mathsf{T}} W_i D)^{-1} D^{\mathsf{T}} W_i \,(\mathbf{y}_i - \tau_{\mathrm{pp}} \mathbf{N}_i^{\mathrm{pp}}/s_i), \qquad \mathrm{Cov}(\hat{\mathbf{a}}_i) = (D^{\mathsf{T}} W_i D)^{-1}, \qquad (W_i)_{cc} = \frac{s_i^2}{\max(N_{ic}, 1)}. \tag{27}$$

All pixels are solved at once with batched normal equations. The pixel-specific pile-up spectrum is subtracted before the solve because it does not fit in a shared design matrix. The median signal-to-noise ratio of the areas per pixel is 204 for As K, 138 for Ga K and 8.0 for In L. Atom-ratio maps follow from Eq. (25) with the converged sensitivities held fixed, so a map costs one matrix product. The errors are propagated from the two areas, treated as uncorrelated.

The weights in Eq. (27) use the observed counts (Neyman's χ^2). This is unbiased for the strong Ga K and As K channels but biases weak peaks low. Channels with fewer counts receive larger weights, and a downward fluctuation pulls the fit down [24, 25]. On peaks of In L intensity the bias reaches tens of per cent per pixel. The In L map should therefore not be interpreted quantitatively pixel by pixel until the weights are taken from the model (Section 15).

12 Resolution stability during the scan

12.1 Diagnostic

The per-pixel solve assumes that the line shapes are the same in every pixel. That is not guaranteed over a 2.68 h scan. To test it, consecutive pixels are summed row by row in acquisition order, and the first two moments of the strongest isolated line are computed in a window of $\pm \min(2.5\,\mathrm{FWHM},\ \frac{1}{2}$ distance to the nearest significant line), here 8842–9661 eV around Ga Kα, after removal of a linear background between the window edges:

$$\bar{E} = \frac{\sum_c n_c E_c}{\sum_c n_c}, \qquad w_{\mathrm{rms}} = \Big(\frac{\sum_c n_c (E_c - \bar{E})^2}{\sum_c n_c} \Big)^{1/2}. \tag{28}$$

These moments do not depend on the normalisation. A step is flagged at row i when the means of the five rows after and before differ by more than six times their noise, estimated robustly from the median absolute successive difference, and by more than 1 eV.

Figure 9a shows a single abrupt step at row 25: the rms width of Ga Kα rises from 87.6 eV to 94.9 eV (8.4 %), with a statistical error of 0.06 eV per row, while the centroid shows no step. Fitting the two segments separately gives 182.2 eV and 194.1 eV FWHM at Ga Kα. The step is not correlated with the count rate (97.7 kcps before, 96.1 kcps after), the dead time or the monitor. The width-change map (Fig. 9b) places it between pixels 5 and 6 of row 25, 3038 s after the start of the scan. Possible causes are a change of a processor setting (e.g. peaking time) or of the detector temperature; the scan metadata do not record either.

12.2 Why it matters, and why window sums do not see it

A width change redistributes counts between overlapping peaks when their shapes are frozen. Ga Kβ (10.26 keV) sits on the low-energy flank of As Kα (10.54 keV), so the Ga/As ratio of a one-shape fit jumps by −1.16 % at the step (Fig. 9a, bottom). The ratio of plain counts in two windows around Ga Kα and As Kα shows no step (0.4639 before, 0.4639 after): a window wide enough to contain the whole peak is insensitive to its width. That is why maps made from window sums look smooth while fitted maps can show such a step.

Table 4: Resolution-tracking options for the per-pixel solve: step of the row-median Ga/As ratio at row 25 (after/before −1), relative spread of the Ga/As map, and median and 5–95 percentile range of the per-pixel χ^2_ν.

option	step (%)	spread (%)	median χ^2_ν	χ^2_ν 5–95 %
`none`	−1.16	1.16	1.079	0.96–1.26
`segments`	0.14	1.03	0.999	0.89–1.11
`linear`	−0.19	1.03	1.009	0.91–1.12
`both`	0.15	1.03	0.998	0.89–1.11

12.3 Correction

Two corrections are implemented, separately or together:

Segments.
The map is cut at the detected steps. Each segment's average spectrum is fitted (Section 9) and its own shape parameters are used in Eq. (27). This is exact for an abrupt change at a row boundary.

Linear width term.
One column is added to D: the derivative of the modelled peaks with respect to a common scaling κ of all Gaussian widths, $d_c = \sum_g \partial \hat{y}_g(c; \kappa\sigma)/\partial\kappa|_{\kappa=1}$ (central difference, $h = 0.02$). Its coefficient δ_i gives the relative width change of the pixel, $\epsilon_i = \delta_i/\lambda_i$, where λ_i projects the pixel's group areas on those of the shape fit. This first-order term handles changes of a few per cent in any pixel, including those inside a row.

Table 4 compares the four options. Either correction removes the step and lowers the median per-pixel χ^2_ν from 1.079 to about 1. Combining them (the default) also corrects the first 6 pixels of row 25, which belong to the new segment but were still measured with the old resolution (width change between −8.5 % and −6.9 %, against −1.4 % for the next pixel).

13 Results on the (In,Ga)As film

13.1 Maps

Figure 10 shows the Ga K and As K areas and the Ga/As atom ratio of the film. The area maps show the horizontal bands of the refills (Section 4.3); the ratio map does not. The ratio map has a median of 0.5032 (0.5026 without resolution tracking) and a pixel-to-pixel standard deviation of 0.0052. The statistical error expected from the area errors, scaled by the square root of the median χ^2_ν, is 0.0044 per pixel. The excess, $\sqrt{0.0052^2 - 0.0044^2} = 0.0027$ (0.0036 without tracking), is real composition variation or residual systematics at the level of half a per cent. Its pattern is not random: the ratio and the Ga map both show a diagonal cross-hatch with a period of a few micrometres. Lattice-mismatched films often show such a composition modulation along misfit-dislocation directions, and it can be compared directly with the strain maps from the Laue patterns of the same scan.

13.2 Uncertainty budget

The statistical error of the average is negligible. The accuracy is set by the model parameters that are not measured. Table 5 and Fig. 11 requantify the same fitted areas after changing one parameter at a time. The Ga/As ratio stays within 0.490–0.514. The largest effect is the KB grazing angle, which is not recorded: 2.5 and 3.5 mrad give 0.5103 and 0.4898, because a steeper angle lowers the high-energy cut-off and with it the excitation of As relative to Ga. Next comes the air path (±2 % for ±2 cm), because the efficiencies of Ga Kα and As Kα are not equal. A beam offset of 25 µrad from the orbit plane gives 0.5011; geometry, film thickness and pressure

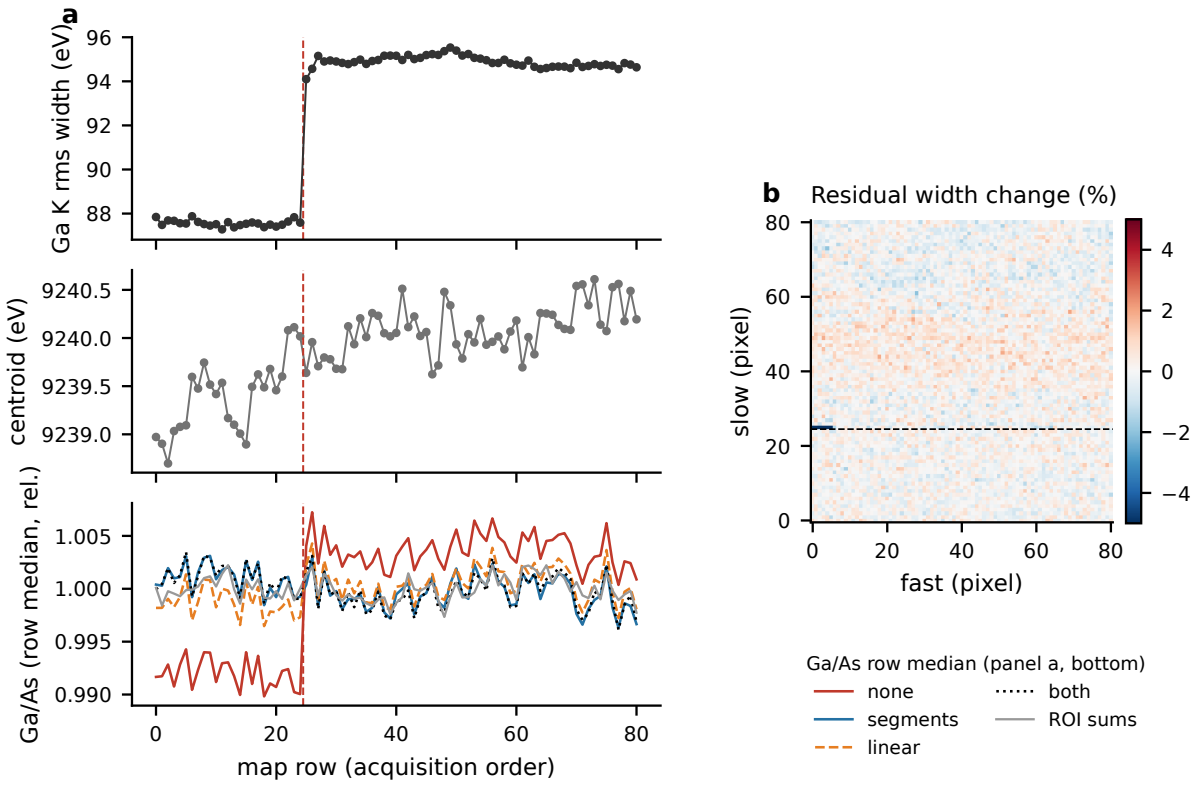


Figure 9: Resolution stability. (a) Per-row rms width and centroid of Ga Kα in acquisition order, and row-median Ga/As area ratio for the four options of Table 4 and for plain window sums (each normalised to its mean); the dashed line marks the detected step. (b) Residual relative width change per pixel from the linear width term, with segments.

contribute a few tenths of a per cent. The vertically integrated source used in earlier versions of the code (0.4825) is listed for reference only; it is not a plausible alternative for this beamline. The In content and the closure are a different matter: the closure ranges from 1.15 to 1.77 for an air path of 18 to 22 cm, and reaching 1 would need 16.5 cm of air with everything else nominal. The measured In L intensity is therefore not a measurement of the In content until the detection path is known; whether the remaining excess comes from the air path alone, from the take-off angle, or from the In L line strengths themselves cannot be decided from this map.

14 Discussion and limitations

What is robust. Ratios of lines of similar energy (Ga K/As K) are robust. Normalisation cancels exactly, the beam spectrum enters through the flux between and above the Ga and As K edges, and the detector and matrix corrections are similar for both lines. The three steps that make a difference at the per-cent level are the bandwidth Jacobian (7.5 %), the vertical acceptance of the beamline (4.5 %) and the sample model (−4.4 %); resolution tracking matters at the level of 1 % locally. What remains uncertain for Ga/As is the incident spectrum itself, chiefly through the KB grazing angles (0.5103–0.4898); a GaAs reference measured in the same configuration would calibrate it directly.

What is not. Soft lines (In L here, and generally L lines of $Z \lesssim 55$ and K lines below 4 keV) depend exponentially on the sample–detector air path. The path is known to a few centimetres only, and its effective length also depends on the take-off angle and on the beam position on the sample. A one-centimetre error changes the In L efficiency by 15 %. The same lines are the most

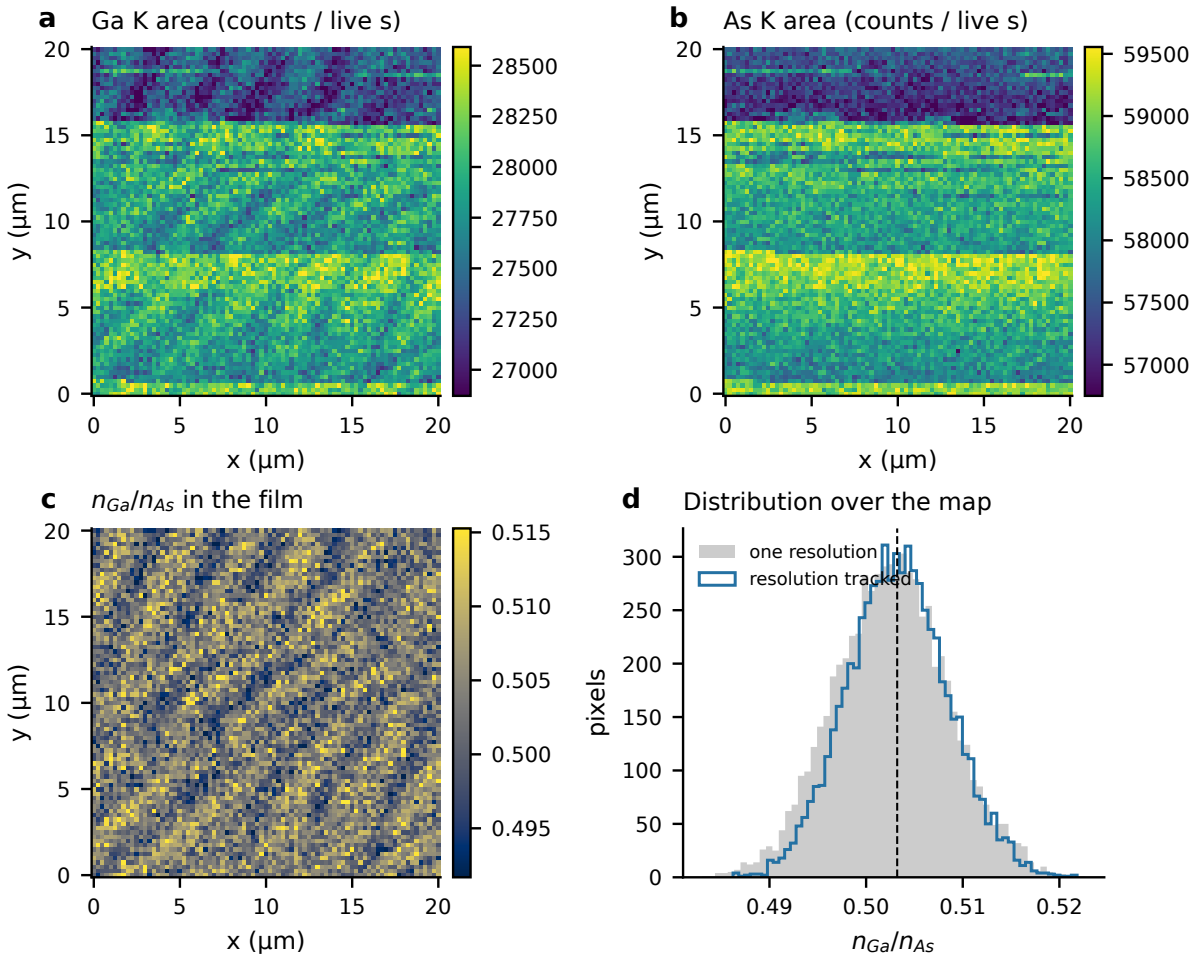


Figure 10: Maps. (a, b) Ga K and As K group areas (all lines, per live second and unit relative monitor). (c) Atom ratio $n_{\mathrm{Ga}}/n_{\mathrm{As}}$ in the (In,Ga)As film with resolution tracking (segments and linear width term). (d) Distribution over the map with and without resolution tracking; the dashed line is the median.

absorbed in the sample and the most affected by the Neyman bias of the per-pixel solve.

Line shape. The largest remaining misfit of the spectrum is the low-energy tail and shelf of the strong lines. The fitted Lorentzian width (11.9 eV) partly mimics it. A tail that is not modelled leaks into whatever lies just below a strong line: here Ga Kα into the Cu K and Ni K region, and As Kα into Ga Kβ.

Monitor. The monitor over-responds at refills, which leaves 2 % steps in all absolute areas (Section 4.3). Ratios are not affected.

Not addressed. Laue Bragg reflections reaching the SDD would appear as narrow, position-dependent peaks at arbitrary energies. None is obvious in the analysis region of this map, but the unidentified 15.8 keV peak may be one. Elastic and Compton scattering of the white beam is part of the smooth background and is described by a polynomial, which is adequate over 10 keV but not over the full range. The incident spectrum is computed, not measured: the KB grazing angles, mirror roughness, slope errors and the exact air density are not constrained by the data, and they mainly affect the high-energy tail of the band and the ratio of elements whose edges are far apart.

Table 5: Sensitivity of the film composition to unmeasured parameters (same fitted areas, requantified). Nominal: 20 cm air at 1013 hPa, 40°/40°, film 1.9 µm, cap 50 nm, on-axis source limited by s3, KB angles 3 mrad. The last four rows change the incident spectrum; the G_1 row is the previous, vertically integrated source model, shown for reference.

case	$n_{\mathrm{Ga}}/n_{\mathrm{As}}$	change (%)	$n_{\mathrm{In}}/n_{\mathrm{As}}$	$(n_{\mathrm{Ga}}+n_{\mathrm{In}})/n_{\mathrm{As}}$
nominal	0.5040	0.0	0.910	1.414
air 18 cm	0.4941	−2.0	0.655	1.149
air 22 cm	0.5142	2.0	1.261	1.775
take-off 30°	0.5067	0.5	0.997	1.503
take-off 50°	0.5027	−0.3	0.861	1.364
film 1.52 µm	0.5061	0.4	0.850	1.356
film 2.28 µm	0.5027	−0.3	0.969	1.471
cap 30 nm	0.5041	0.0	0.926	1.430
cap 70 nm	0.5040	0.0	0.895	1.399
no cap	0.5041	0.0	0.949	1.453
990 hPa	0.5017	−0.5	0.844	1.346
source G_1 (all ψ)	0.4825	−4.3	0.834	1.316
beam offset 25 µrad	0.5011	−0.6	0.901	1.402
KB angles 2.5 mrad	0.5103	1.2	0.896	1.406
KB angles 3.5 mrad	0.4898	−2.8	0.880	1.369

15 Recommendations

15.1 Instrument and acquisition

1. **Measure the detection geometry for every experiment.** Record the sample–detector distance (to ±2 mm), the take-off angle and the incidence angle, and store them in the scan metadata. Alternatively, flush the detection path with He or use a vacuum snout, which brings the In L efficiency from a few per cent to near unity and removes the dominant uncertainty of soft lines.

2. **Record the beam-defining geometry.** Store the KB grazing angles and the units of the slit gaps with the scan, and check that s3 is centred on the beam (an offset of 50 µrad from the orbit plane changes the Ga/As sensitivity by −2.67 %). The spectrum model depends on them, whereas the exact gaps matter little as long as the vertical acceptance stays below about 20 µrad.

3. **Record the local pressure and temperature** (or take them from the ESRF logging), since the In Lα efficiency through 20 cm of air changes by 8 % between 990 and 1013 hPa.

4. **Characterise the monitor.** Measure its dark offset and linearity against the ring current, or log the ring current with every point and model the monitor as $m = a + bI$; consider a monitor after the KB mirrors.

5. **Fix and record the processor settings** (peaking time, gap, thresholds) and the detector temperature for the whole scan; this would have avoided the resolution step of Section 12.

6. **Extend the MCA range** to about 30 keV. The upper part of the band excites the K lines of In (24.2 keV) and the pile-up spectrum is fully recorded, which allows an independent In quantification free of the air-path problem (but sensitive to the mirror cut-off).

7. **Keep the input rate moderate** (or use a shorter count time per point at lower rate) where dead time exceeds 30–40 %. Pile-up corrections are first order in the rate.

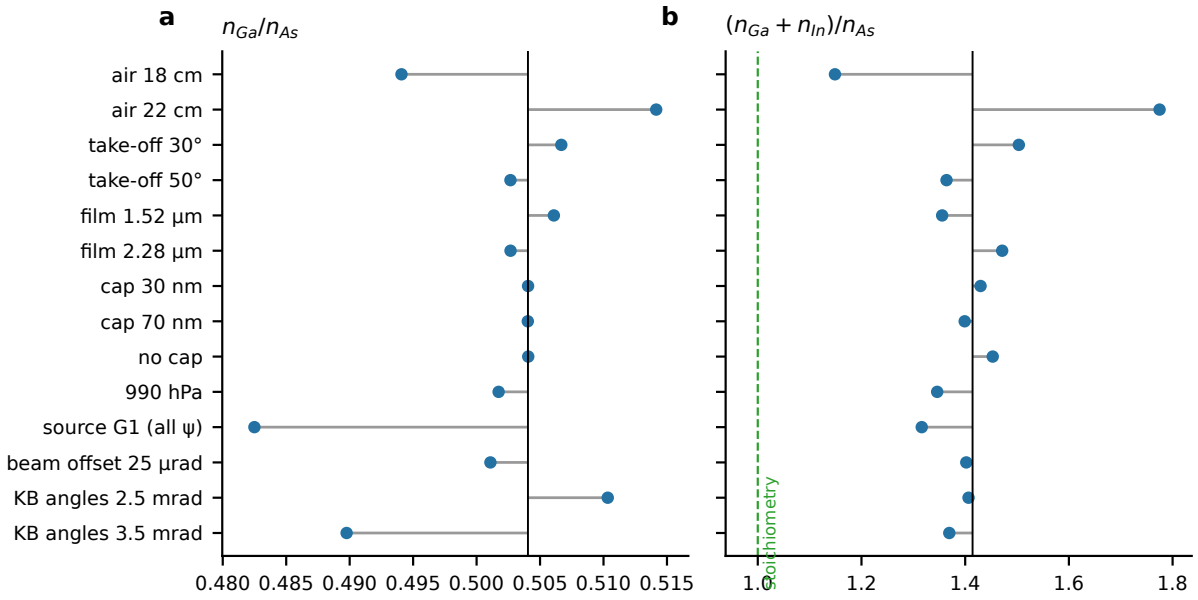


Figure 11: Sensitivity of (a) the Ga/As ratio and (b) the closure $(n_{\mathrm{Ga}}+n_{\mathrm{In}})/n_{\mathrm{As}}$ to the cases of Table 5; the vertical line is the nominal result and the dashed line stoichiometry.

15.2 Calibration with standards

8. **Measure binary reference materials** in the same geometry: GaAs, InP and InAs wafers (bulk) calibrate the Ga/As and In/As (and In/P) sensitivities directly, including the effective air path and detector efficiency. A GaAs wafer would turn the closure test into a calibration of the In L line.

9. **Measure a thin-film standard** (e.g. a certified multi-element thin film, or an $\mathrm{In}_x\mathrm{Ga}_{1-x}\mathrm{As}$ layer of known x and thickness from XRD or RBS) to test the thin-film matrix model separately from bulk absorption.

10. **Validate the incident spectrum** with pure-element foils of increasing atomic number (e.g. Ti, Cu, Zr, Ag, In, Sn): the Kα yields of foils of known thickness map the spectral density across the band, and a fit of the mirror angles, roughness and air path to them replaces the computed spectrum by a measured one.

15.3 Analysis

11. **Fit the energy calibration** (gain and offset) on the average spectrum, or at least check it against the strong lines, rather than holding the beamline calibration fixed.

12. **Add a low-energy tail and shelf** to the line shape (Hypermet-type step and exponential tail [22], with amplitudes as smooth functions of energy). This is the largest remaining misfit, and it biases peaks lying just below strong lines.

13. **Use model-based (Pearson) weights in the per-pixel solve**, $W_{cc} = s_i^2/\hat{N}_{ic}$, obtained by one or two re-weighting iterations starting from Eq. (27), or a Poisson likelihood [24]; this removes the downward bias on weak peaks such as In L.

14. **Treat Laue reflections explicitly**: detect narrow peaks that are not fluorescence lines from the per-pixel residuals, and mask or model them pixel by pixel.

15. **Compute pile-up from the input rate** when the dead time is high, and allow the resolving time to depend on the energies of the two pulses.

16. **Propagate the uncertainty budget**: report, with every map, the statistical error and the systematic range from Table 5-type requantifications using the actual uncertainties of geometry, air path and thicknesses.

17. **Use the closure as a consistency check, not as a constraint**. Fitting the air path to force stoichiometry makes the In content agree by construction and hides any other error in the soft-line chain. It is legitimate only once standards have shown that the remaining model is right.

A Software

The chain is implemented in three modules at the root of the `pyXRT` repository; the notebook `fluo_analysis_clean.ipynb` runs it with per-sample settings, and `make_figures.py` in this folder reproduces this document. Table 6 maps the sections to the functions.

Table 6: Where each step lives.

step	functions
reading	`bliss_hdf5_reader.BlissMeshScan`
normalisation (Sec. 4)	`xrf_fit.normalise_spectra`, `average_spectrum`, `counting_error`
incident beam (Sec. 5)	`xrf_beamline.bm32_from_h5`, `BM32_WHITE_BEAM`, `Beamline` (`spectrum`, `acceptance`, `describe`), `BendingMagnetSource.angular_flux`, `Slit`, `Mirror`, `per_unit_energy`
detector (Sec. 6)	`xrf_fit.Detector` (`from_fwhm`, `sigma_ev`, `efficiency`, `escape_lines`), `pileup_spectra`, `average_pileup_template`
line groups (Sec. 7)	`xrf_fit.element_line_groups`
sample (Sec. 8)	`xrf_matrix.Geometry`, `Layer`, `Stack`, `MatrixModel`
average fit (Sec. 9)	`xrf_fit.XRFModel`, `fit_average_spectrum`
quantification (Sec. 10)	`xrf_matrix.quantify_layer`, `atom_ratio`; `xrf_fit.number_ratio`
maps (Sec. 11)	`xrf_fit.fit_map`, `MapResult`
resolution (Sec. 12)	`xrf_fit.width_window`, `peak_width_profile`, `detect_steps`, `segment_labels`, `fit_segments`; `fit_map(segments=, width_correction=)`

Reproducing this document. From the repository root, with `InPoSi_v2_center.h5` present:

```
python docs/xrf_quantification/make_figures.py  # figures, results.json (3 min)
python docs/xrf_quantification/make_macros.py   # results_macros.tex
cd docs/xrf_quantification && tectonic xrf_quantification.tex
```

The first command runs the complete chain; the second turns its output into the `\r...` macros used for every number in the text.

Main settings. Beam from `bm32_from_h5` (recorded slit gaps, mirror angles and radii, ring current; KB angles 3 mrad, angle spread on); energy grid 3–40 keV in 10 eV steps; elements In, Ga, As, Cu, Ni, Ca, Ar (K and L families); minimum relative line intensity 0.005; analysis region 2.9–12.6 keV, background order 3; pile-up region 2.9–20.3 keV, background order 5; initial resolution 135 eV at 5.9 keV with $F = 0.12$; secondary fluorescence on a 40 eV grid with 160 direction points; step detection window 5 rows, threshold 6, minimum step 1 eV.

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